

MATH5320 – Homework VII

1. Merton

Assume a Merton model for default with a constant risk free rate of r and an initial firm value of \$1100000. The firm issued a zero coupon bond with face value B that matures at time T :

$$\begin{aligned}dV &= \mu V dt + \sigma V dW \\ \mu &= 0.023 \\ \sigma &= 0.280 \\ r &= 0.055 \\ V_0 &= 1100000 \\ T &= 5.0 \\ B &= 850000\end{aligned}$$

(a) What are the real world and risk neutral (with respect to the money market account as numéraire) probabilities that the firm will default within 5.0 years?

(b) What is the probability that the firm will default in between 3.0 and 4.0 years?

1)

Merton Model refers to a structural model, that generally assumes that a firm's asset value follows a process and defaults if the asset value gets too low relative to the debt, i.e. $V_t < D$. It's called structural because it is based off a balance-sheet structure, and also, under the basic Merton model, there is no default before maturity (at T), and at $t=0$, the instantaneous default probability is zero. It models the assets like the following:

$$dV = \mu V dt + \sigma V dW$$

Recall from 5264, the solution to that is:

$$V_T = V_0 \exp((\mu - \sigma^2/2)T + \sigma W_T)$$

And the only debt that is modelled is through a ZCB of value B , and is only checked for default at T . IT functions effectively in a way where equity is a call option on the assets with strike B , and the debt is such that it is ZCB – put on the assets.

a)

For this question, we are evaluating under the historical measure, P , the probability that the firm will default within 5 years.

Merton default probabilities

Merton default probability at time t :

$$\begin{aligned}p_t(T) &= Q(V_T < B) = E^Q[1_{V_T < B}] \\ V_t - B &= \text{VaR}^Q(V, T, 1 - p_t(T))\end{aligned}$$

So we see that default probabilities in the Merton model are just one minus the percentiles for the VaR that brings the asset value down to the level of the debt at the debt horizon date.

So we get (from the derivation of GBM VaR):

$$\begin{aligned}p_t(T) &= Q(V_T < B) \\ &= \Phi\left(\frac{\log\left(\frac{B}{V_t}\right) - \left(r - \frac{\sigma^2}{2}\right)(T - t)}{\sigma\sqrt{T - t}}\right)\end{aligned}$$

Because it is the Merton model, and “we default” within 5 years and “default at $T = 5$ ”, it is the same event such that

$$P(\text{default before } T) = P(\text{default at } T) = P(V_T < B) = \Phi(-d_2)$$

In this case, unlike the reduced form model, there is no hazard rate or survival curve to integrate.

$\Phi(-d_2)$ is literally just the probability that we end up below B , or in other words “default”. As such, the survival rate is basically:

$$s_0(u) = 1 - \Phi(-d_2) \cdot 1_{u \geq T}$$

Now going back to the formula, for the probability for default, with two candidate drift terms, that is r or μ

Under Q (RN measure) ($drift - term = r = 0.055$):

$$d_2 = \frac{0.2578 + (0.055 - 0.0392) \times 5}{0.6261} = \frac{0.2578 + 0.0790}{0.6261} = \frac{0.3368}{0.6261} = 0.5380$$

$$p_{RN} = \Phi(-0.5380) = 29.53\% \text{ (2dp)}$$

Under P (historical measure) ($drift - term = \mu = 0.023$):

$$d_2 = \frac{0.2578 + (0.023 - 0.0392) \times 5}{0.6261} = \frac{0.2578 - 0.0810}{0.6261} = \frac{0.1768}{0.6261} = 0.2824$$

$$p_{RW} = \Phi(-0.2824) = 38.88\% \text{ (2dp)}$$

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log(V0/B) = ln(1,100,000/850,000) = 0.257829
sigma*sqrtT = 0.28 * sqrt(5.0) = 0.626099
sigma^2/2 = 0.039200

Risk-Neutral (nu = r = 0.055)
d2 = [0.2578 + (0.0550-0.0392)*5.0] / 0.6261 = 0.537981
p = Phi(-0.5380) = 0.295295 (29.53%)

Real-World (nu = mu = 0.023)
d2 = [0.2578 + (0.0230-0.0392)*5.0] / 0.6261 = 0.282430
p = Phi(-0.2824) = 0.388807 (38.88%)

Summary:
Risk-Neutral DP : 29.53% (pricing)
Real-World DP : 38.88% (risk management)
Difference : 9.35 pp
p_RW > p_RN because mu (0.023) < r (0.055): assets grow
more slowly in reality => higher chance of falling below B.
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So we note that under P, we have a higher probability of default, and we note that every option pricing model including Merton, is derived under the risk-neutral measure Q. And under Q, the discounted asset is a martingale. So, under Q, we are basically looking at the probability of default, when assets are martingales, and then “value” what something is worth today”.

But in the real-world, assets grow at drift, which is much, much lower than the r (*in this case*). As it is lower, then it will be more likely to drift below B, so the IRL or P-measure default probability is higher. As everything else is the same, the difference in default probability is purely driven by the different drift term.

b)

Building on top of what we had before, now we want the “between three to four years”. We note that, now, this follows on the critical limitation of Merton model, where there is no possibility of default before time T. The idea is that default is only defined at maturity, so,

The model defines default only at the debt maturity date T . So for any interval that does not contain T , where τ is default time:

$$P(\tau \in (x, x + 1]) = 0.$$

Following on top of this, there are three cases such that:

There are three general cases:

If $x + 1 < T$, then

$$P(\tau \in (x, x + 1]) = 0.$$

If the interval contains T , meaning $x < T \leq x + 1$, then

$$P(\tau \in (x, x + 1]) = P(V_T < D).$$

If $x \geq T$, then

$$P(\tau \in (x, x + 1]) = 0,$$

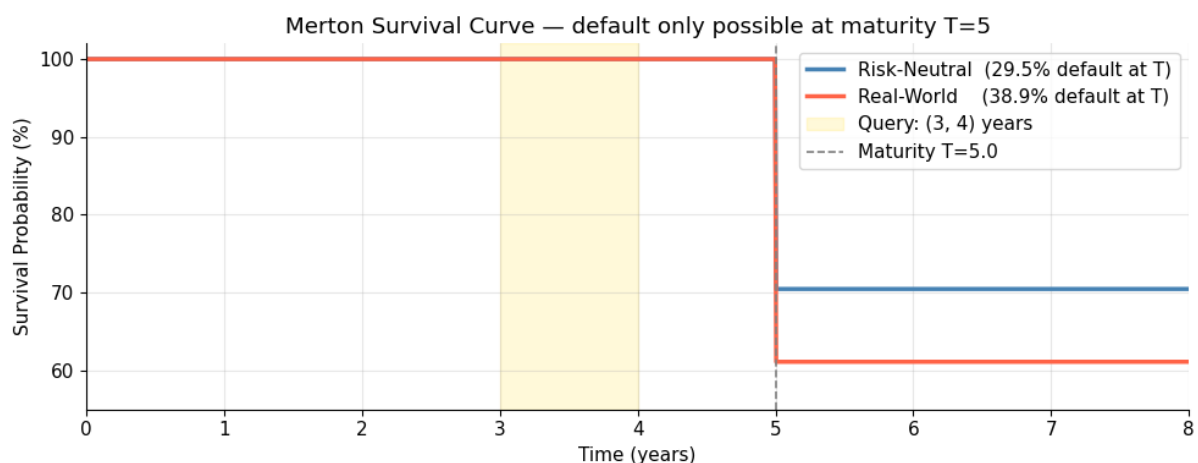
because default, if it happens, already happened at T , and there is no second default later for that bond. And we note that in this model the default time is such that: $\tau = T$.

So then, we note that the $P(\text{default} \in (3,4)) = s_0(3) - s_0(4) = 1 - 1 = 0$

This is because, if we look at the Merton Survival curve, before T , we have the $s_0(3) = 1$. And then, in this case, it will literally be $1 - 1 = 0$ (as shown above). If we were to look at the curves from the step functions amalgamated, we will have:

Interval	s0(start)	s0(end)	P(default in interval)
(0,1)	1	1	0%
(1,2)	1	1	0%
(2,3)	1	1	0%
(3,4)	1	1	0%

But then, it is at $T = 5$, where there is default probability, as shown before.



This is because, we defined $T = 5$ prior!

This reflects the general idea that Merton default is predictable, such that as we approach T , we know if we are going to default, by look at the asset value. And the practical implication of this is that if we tried to price a CDS on this using Merton, we will get a spread of 0 (over RFR), as the model will say that there is no probability of default before T . This is because we are either exactly at T (where it can default), or it never happens at all. This is a fairly weak point, and has many unrealistic pricing implications so Black-Cox was introduced.

2. Historical VaR, relative changes

Continuing with the settings in the previous homeworks, namely A being AAPL, C being CAT, and P being the portfolio consisting of 24679 shares of AAPL, and 171 shares of CAT.

Compute the 99% 5 day historical VaR and the 97.5% 5 day historical ES for a \$10,000 investment in A , C , and P on each day in the last 20 years. Do this for each date d by applying the previous 5 years of daily log returns to the position on that date (windowed).

How do the historical VaR and ES compare to the previous results?

Would it matter if the VaR for P was computed from the historical time series changes for P or from the historical time series changes for A and C applied to the underlying stocks in the portfolio? Why or why not?

2)

The problem asks us to compute the 99% 5d historical VaR and the 97.5% 5d historical ES on the a 10,000 position for the last 20 years, and use log changes, s.t. we note that a rolling 5-year window of daily data, as the foundation for later calculations. For each evaluation date d , we take the previous 5 years of daily data, construct overlapping 5-day historical log-return scenarios, map each scenario into a 5-day P&L on a \$10,000 position, and then compute the empirical 99 percent VaR and 97.5 percent ES from that historical scenario distribution

For hw07 problem we have the following results table.

Metric	Latest	20yr Avg	Peak	Peak Date
VaR_A	\$ 905.78	\$ 1136.17	\$ 1532.91	2008-11-21
ES_A	\$ 964.02	\$ 1134.31	\$ 1544.15	2009-02-24
VaR_C	\$ 969.13	\$ 1122.61	\$ 1600.45	2011-08-09
ES_C	\$ 1026.20	\$ 1160.09	\$ 1568.80	2011-09-26
VaR_P_direct	\$ 898.76	\$ 1077.93	\$ 1448.06	2008-11-21
ES_P_direct	\$ 958.00	\$ 1088.05	\$ 1470.13	2009-02-24
VaR_P_from_AC	\$ 894.17	\$ 1100.88	\$ 1479.57	2009-07-10 *
ES_P_from_AC	\$ 952.41	\$ 1104.72	\$ 1510.32	2009-07-10 *
* = built from A & C log-return scenarios separately				

Note we did it by scaling the 1-day returns by $\sqrt{5}$ previously, and obtained similar results:

Metric	Latest	20yr Avg	Peak	Peak Date
VaR_A	\$ 923.30	\$ 1130.15	\$ 1422.60	2008-12-18
ES_A	\$ 984.78	\$ 1194.33	\$ 1530.63	2006-02-13
VaR_C	\$ 1001.93	\$ 1123.29	\$ 1440.88	2011-09-23
ES_C	\$ 1057.63	\$ 1185.59	\$ 1518.07	2011-09-23
VaR_P_direct	\$ 914.89	\$ 1085.94	\$ 1368.52	2008-11-21
ES_P_direct	\$ 978.61	\$ 1142.73	\$ 1400.42	2009-01-21
VaR_P_from_AC	\$ 906.80	\$ 1101.53	\$ 1380.69	2009-07-10 *
ES_P_from_AC	\$ 972.82	\$ 1163.31	\$ 1449.47	2009-07-10 *
* = built from A & C log-return scenarios separately				

One notable result is that CAT's latest 5-day historical ES exceeds AAPL's, at \$1,026.20 versus \$964.02, which reverses the ordering seen in some earlier parametric results in HW5. This reflects CAT having more extreme log-return scenarios in the 5yr window relative to its recent volatility level, which the parametric model's σ^2 estimate smooths over. AAPL and portfolio peaks cluster around November to December 2008, consistent with the acute phase of the GFC dominating those windows.

We should note that direct from VaR difference, **does slightly differ, and does matter**, as we note that the two methods are not equivalent. We note that the portfolio VaR difference (direct minus from A&C) is nonzero: mean = -\$22.95, standard deviation = \$39.84, and maximum absolute difference = \$225.75. The two methods are not equivalent for log returns because:

$$\log(w_A \cdot A + w_C \cdot C) \neq w_A \cdot \log(A) + w_C \cdot \log(C)$$

by Jensen's inequality (log is concave). The portfolio log-return is nonlinear in the component log-returns, so applying weights to individual scenarios and summing gives a different distribution than directly using the portfolio return series. The difference is modest on average but can still reach \$225.75 on individual dates, so it is not negligible for risk measurement

We note that the difference is small, but in absolute cases, it can reach up to almost 226, which does make a difference for risk assessment.

Does it matter (relative $\angle$ log changes)? – maybe?????

VaR diff: mean=-22.95 std=39.84 max_abs=225.75

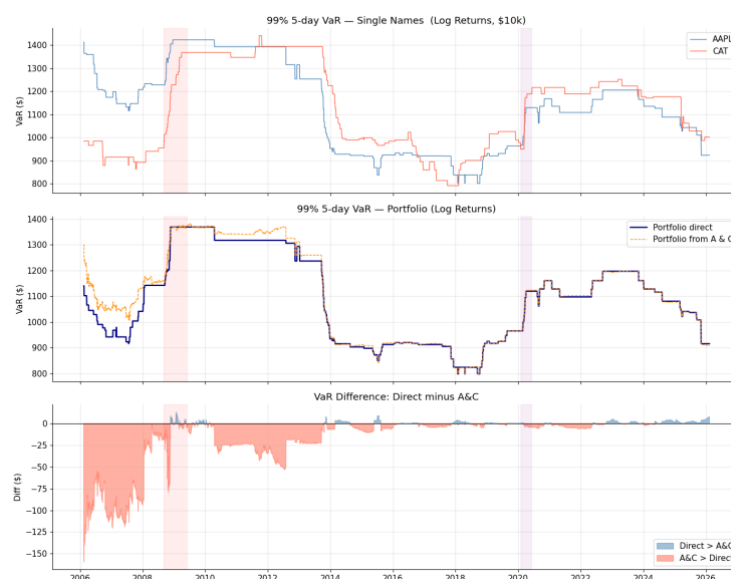
ES diff: mean=-16.67 std=26.85 max_abs=129.55

Reason: $\log(w_A \cdot A + w_C \cdot C) \neq w_A \cdot \log(A) + w_C \cdot \log(C)$ [Jensen's inequality]

Portfolio log-return is nonlinear in component log-returns.

This was also noted in MATH5380, when attributing returns, etc. the portfolio and its individual attributes **do not line up**, when using log-returns, and as such, matters here in performance attribution more broadly.

As shown below the difference can change a lot across the back testing period, especially during 2006, when $A \& C > \text{Direct}$.



We need to note that previously, in HKWK5 Q3, we used the parametric GBM position on individual \$10K positions, using 2y5y10y rolling windows on their equivalent EWMA values. As shown below, these gave about 831-963\$ for AAPL, and about 869-936\$ for CAT. In HW6 Q2 (methodology detailed later) we extended this parametrically to the full portfolio using a 2yr window, giving a GBM VaR of ~\$836 per \$10k equivalent. The portfolio VaR per 10k (\$836) fell below either single-stock HW5 figure, reflecting the diversification benefit from combining AAPL and CAT with rolling correlation $\rho = 0.35$, so combining them reduces portfolio variance below the weighted average of the individual variances through the cross term $2w_Aw_C\rho\sigma_A\sigma_C < 2w_Aw_C\sigma_A\sigma_C$.

Noting previous weeks, for hw05 Q3, as shown:

Time-average VaR/ES over the last 20y (NaNs skipped):							
Horizon		10y		2y		5y	
Method		exp	window	exp	window	exp	window
Asset Measure							
AAPL	ES	1103.88	1187.36	971.02	978.06	1003.98	1023.33
	VaR	1098.20	1181.28	966.01	973.01	998.79	1018.03
CAT	ES	1025.61	1039.73	987.14	989.36	1002.08	1005.33
	VaR	1020.49	1034.54	982.20	984.41	997.07	1000.30

Using the corrected historical 5-day methodology, the latest historical VaR is \$905.78 for AAPL, \$969.13 for CAT, and \$898.76 for the portfolio, all meaningfully higher than every parametric equivalent. The gap versus the closest parametric AAPL figure of about \$843 is roughly \$62.78, and versus the portfolio parametric figure of about \$836 is roughly \$62.76.. This is a fundamental structural difference, as the parametric model fits a lognormal to the 2y window, and then reads the tail at the 1st percentile, where it is smoothed off, whereas historical simulation instead ranks all 1,260 scenarios and takes the 13th worst directly, meaning days like Liberation Day (3 April 2025), where AAPL's log-return was -9.70%, enter the VaR at full weight rather than being absorbed into a fitted estimate.

For hw06 Q2, we have:

Q2 · VaR & ES – Latest Values	
Method 1 – GBM (exact lognormal, Q1 μ/σ^2):	
VaR (99%, 5d) :	\$ 579,445
ES (97.5%) :	\$ 662,028
Method 2 – Two-Stock Normal Portfolio:	
VaR (99%, 5d) :	\$ 597,001
ES (97.5%) :	\$ 600,071
BM Driver Corr $\rho(dw_A, dw_C)$: 0.3507	

Here, we used the estimates from the 2y5y10y windows, and corresponding estimated exponential weights. For the portfolio (24,679 AAPL + 171 CAT), we compute a 5-day 99% VaR and 97.5% ES on the actual dollar position using a 2-year rolling window for parameter estimation. Two methods are used, for the GBM exact which is lognormal, who Two-stock normal portfolio, and applies the normal VaR/ES formula. Now it is no longer normalised to 10k. =Normalising back to a \$10k equivalent by dividing by portfolio value and rescaling:

\$579,445 / \$6,931,590 × \$10,000 = \$836,

\$662,028 / \$6,931,590 × \$10,000 = \$955,

\$597,001 / \$6,931,590 × \$10,000 = \$861,

\$600,071 / \$6,931,590 × \$10,000 = \$866.

This is equivalent to \$836 and \$861 per \$10k respectively. Refer to the above results.

Method	Source	VaR / \$10k	ES / \$10k
Parametric GBM (2yr window)	HW5 AAPL	\$843	\$963
Parametric GBM (2yr window)	HW5 CAT	\$870	\$1,000
GBM lognormal (portfolio)	HW6 Q2	\$836	\$955
Two-stock Normal (portfolio)	HW6 Q2	\$861	\$866
Historical simulation (AAPL)	HW7 Q2	\$923	\$985
Historical simulation (CAT)	HW7 Q2	\$1,002	\$1,058
Historical simulation (portfolio)	HW7 Q2	\$915	\$979

Note: these results comparison, were with the old scaled sqrt(5) approach, but conclusion remains similar

Notably, the normal l ES (\$866) is the single lowest ES figure in the entire table, lower even than the HW5 single-stock AAPL parametric ES (\$963), which makes it the clearest evidence that normality is the binding assumption doing the most damage in terms of risk assessment. Notably, the ES/VaR ratio of 1.005 is economically implausible, and it says the average loss in the worst 2.5% of scenarios is almost identical to the loss at the 1% threshold, which means that it only happens as the normal tail is **far too thin**. GBM gives 1.142, and the historical simulation gives 1.070, and as such, it does showcase the asymmetry and fat-tailedness of equity returns.

Over the 20-year backtest, the average historical 5-day VaR is \$1,136.17 for AAPL and \$1,077.93 for the portfolio. It is higher than the short-window parametric averages (approx. \$973 AAPL, approx. \$984 CAT) because the historical method captures the full severity of 2008 to 09 GFC scenarios that a fitted normal distribution compresses, and therefore obscures.

3. Historical VaR, absolute changes

Repeat the previous problem applying absolute changes instead of log changes. How do the VaR and ES change? In this case, would it matter if the calculation for P is done using the historical changes for P or using the historical changes for A and C applied to the underlying stocks in the portfolio? Why or why not?

Now, we look at the previous problem which was done with log-returns, and now we note the following differences. Generally, log-returns are preferred as they are additive and are far easier for calculations, and looking below, in terms of absolute and dollar changes for VaR and ES calculations, it does not matter much in terms of differences. For this dataset, the choice between relative/log changes and absolute/dollar changes **does matter quantitatively**, the portfolio VaR is \$898.76 under relative changes versus \$572.27 under absolute changes on the latest date, and the average difference over the sample is also substantial. It does **not matter** whether the portfolio VaR is computed directly from the portfolio price series or reconstructed from AAPL and CAT price changes, because the portfolio dollar change satisfies:

$$\Delta_P = N_{AAPL} * \Delta_A + N_{CAT} * \Delta_C$$

This is linear in the component changes, so constructing the portfolio scenario-by-scenario from A and C gives an algebraically identical result to using the portfolio price series directly. This contrasts with Q2, where the analogous VaR difference had mean - \$22.95 and maximum absolute difference \$225.75, because log-returns do not satisfy a linear portfolio identity:

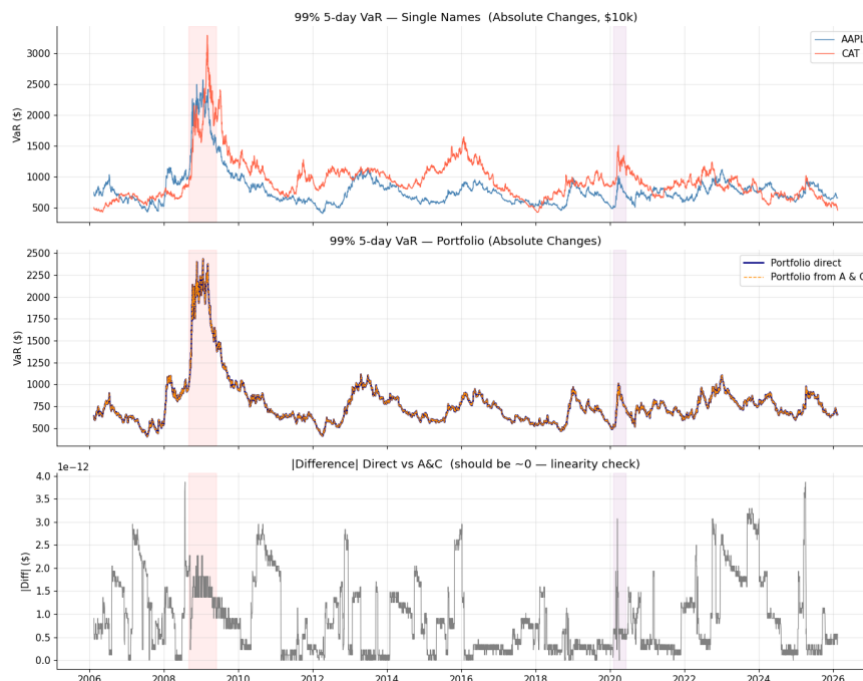
$$r_P \neq N_{AAPL} * r_A + N_{CAT} * r_C$$

The correct first-order approximation is $r_P \approx \sum w_i r_i$, and needs to account for Jensen's inequality correction.

Metric	Latest	20yr Avg	Peak	Peak Date
VaR_A	\$ 579.54	\$ 784.90	\$ 2910.56	2009-01-20
ES_A	\$ 676.07	\$ 790.10	\$ 2803.78	2009-01-20
VaR_C	\$ 393.16	\$ 957.13	\$ 3543.44	2009-03-02
ES_C	\$ 411.33	\$ 978.68	\$ 3573.38	2009-03-02
VaR_P_direct	\$ 572.27	\$ 763.11	\$ 2708.82	2009-01-20
ES_P_direct	\$ 664.57	\$ 769.54	\$ 2645.11	2009-01-20
VaR_P_from_AC	\$ 572.27	\$ 763.11	\$ 2708.82	2009-01-20 *
ES_P_from_AC	\$ 664.57	\$ 769.54	\$ 2645.11	2009-01-20 *

* = portfolio P&L built from A & C absolute changes separately

It is worth noting that CAT's absolute-change VaR of \$393.16 is much lower than its relative/log-return VaR of \$969.13, a gap of \$575.97. For AAPL, the corresponding gap is smaller but still material: \$905.78 versus \$579.54, a difference of \$326.24. CAT also has the larger peak absolute-change VaR, at \$3,543.44 versus \$2,910.56 for AAPL, the reverse of the log-return ordering.



For the portfolio under absolute changes, the latest direct VaR is \$572.27 and the latest ES is \$664.57, while the 20-year averages are \$763.11 for VaR and \$769.54 for ES. Compared with the relative/log-return approach, the portfolio VaR is lower under absolute changes by \$326.49 on the latest date and by \$314.82 on average over the sample. For ES, the corresponding gaps are \$293.43 on the latest date and \$318.51 on average. This is because the log-return method applies multiplicative shocks proportional to current value, whereas the absolute-change method applies additive dollar shocks

But, if we compare the direct portfolio calculation with the portfolio rebuilt from AAPL and CAT absolute changes, the maximum absolute difference is only 2.16×10^{-12} for VaR and 9.09×10^{-13} for ES, which is just floating-point noise. We note that also, every since of the approx.. 5000 rows agree to within 10^{-8} (done in old workbook), and it is basically 2x machine epsilon, such that it accumulates it in a way that differences are technically there.

Looking t it in terms of maximum dollar terms, we get the following:

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Does it matter (absolute  $\Delta$  dollar changes)? NO
Max |VaR difference|: 2.16e-12 (floating-point noise only)
Max |ES difference|: 9.09e-13

Reason:  $\Delta_P = N\_AAPL \cdot \Delta_A + N\_CAT \cdot \Delta_C$  (exact linearity)
Both methods are algebraically identical.
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The idea is that the absolute Δ is the same as its components, and its linearity is exact, whereas for log-returns, this relationship falls apart, as non-linearity introduces the Jensen's Inequality correction.



Over a long sample where equity prices trend upward, multiplicative return shocks generally produce larger tail losses than dollar-change shocks. The time-series plots illustrate this behaviour. The log-return VaR and ES (blue) remain persistently above the absolute-change measures (red) for most of the sample, with the largest divergence occurring during periods of elevated volatility such as the 2008–2009 financial crisis. This is in direct contrast to the previous q, where portfolio reconstruction produced different results due to the non-linearity of log-returns, the absolute-change framework preserves the exact linear relationship between the portfolio and its components.

Notably, the portfolio VaR and ES under absolute changes are substantially smaller than under the log-return approach.

Metric	Rel avg		Abs avg		Rel latest		Abs latest	
VaR Portfolio	\$	1085.94	\$	772.33	\$	914.89	\$	654.52
ES Portfolio	\$	1142.73	\$	814.55	\$	978.61	\$	686.39

This is because, as mentioned already, that the log-return method has multiplicative shocks applied to portfolio value, whereas absolute change method has the additive dollar shocks being independent of the current price level. Over a longer sample, where equity prices trend upward, multiplicative return shocks generally produce larger tail losses than dollar-change shocks.

Therefore, while the choice between log-returns and absolute changes materially affects the magnitude of VaR and ES, the method used to construct the portfolio from its components does not

matter under absolute changes because the portfolio P&L is exactly linear in the underlying price changes.